02

Information Systems in Organizations



Principles

The use of information systems to add value to the organization is strongly influenced by organizational structure and the organization's attitude and ability to change.

Because information systems are so important, businesses need to be sure that improvements to existing systems or completely new systems help lower costs, increase profits, improve service or achieve a competitive advantage.

Cooperation between business managers and IS personnel is the key to unlocking the potential of any new or modified system.

Learning Objectives

- Identify the value-adding processes in the supply chain and describe the role of information systems within them.
- Provide a clear definition of 'organizational structure' and 'organizational change' and discuss how these affect the implementation of information systems.
- Identify some of the strategies employed to lower costs or improve service.
- Define the term 'competitive advantage' and discuss how organizations are using information systems to achieve such an advantage.
- Discuss how organizations justify the need for information systems.
- Define the types of roles, functions and careers available in information systems.

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Why Learn About Information Systems in Organizations?

The impact that computers have in organizations cannot be overstated. Office work has been transformed almost beyond all recognition, and many workers could not operate without their computer. All of this happened before the rise in popularity of the Internet as a channel for sharing information. When that happened, the Internet changed everything all over again! No matter what path your career takes, you will almost certainly come into contact with information systems every day. Marketing departments, accounts departments, order processing, shipping and logistics all rely on information systems. Researchers, medical doctors, mechanics – it is difficult to think of a profession where the computer does not play a central role. Even musicians use information systems to get the sound they want. In this chapter, you will see how the use of information systems in every part of organizations can help produce higher-quality products and increase their returns on investment.

2.1 An Introduction to Organizations

organization A formal collection of people and other resources established to accomplish a set of goals.

An **organization** is a formal collection of people and other resources established to accomplish a set of goals. The primary goal of a for-profit organization is to maximize shareholder value, often measured by the price of the company stock. Non-profit organizations include social groups, religious groups, universities, charities and other organizations that do not have profit as their goal.

An organization is a system, which, as you will recall from Chapter 1, means that it has inputs, processing mechanisms, outputs and feedback. Resources such as materials, people and money serve as inputs to the organizational system from the environment, go through a transformation mechanism and then are produced as outputs to the environment. The outputs from the transformation mechanism are usually goods or services which are of higher relative value than the inputs alone. Through adding value or worth, organizations attempt to achieve their goals.

How does the organizational system increase the value of resources? In the transformation mechanism, subsystems contain processes that help turn inputs into goods or services of increasing value. These processes increase the relative worth of the combined inputs on their way to becoming final outputs. Consider a car maker. Its inputs are the staff it has hired, the assembly equipment it has bought, raw materials such as metal and plastic, and pre-assembled components such car radios. The processing that it does is turning the materials into finished vehicles, which are the output. The finished product is worth more than the cost of the components. This amount is the value that has been added.

value chain A series (chain) of activities that includes inbound logistics, warehouse and storage, production, finished product storage, outbound logistics, marketing and sales, and customer service.

The **value chain**, popularized by Michael Porter in his book, *Competitive Strategy*,[1] is a useful tool for analyzing where and how this value gets added. The value chain is a series (chain) of activities that includes inbound logistics, warehouse and storage, production, finished product storage, outbound logistics, marketing and sales, and customer service. The value chain of a manufacturing company is shown in Figure 2.1.

Analyzing value chains when developing information systems often results in efficient transaction processing systems (explained fully in a later chapter), an expanding market and the sharing of information.[2] The value chain is used to examine what happens to raw materials to add value to them before the finished product is sold to customers. Information systems can be focused on those activities that add the most value. The value chain can also reveal linkages between different activites (say marketing and production) which can be exploited using IS (to increase communication between the two for instance).

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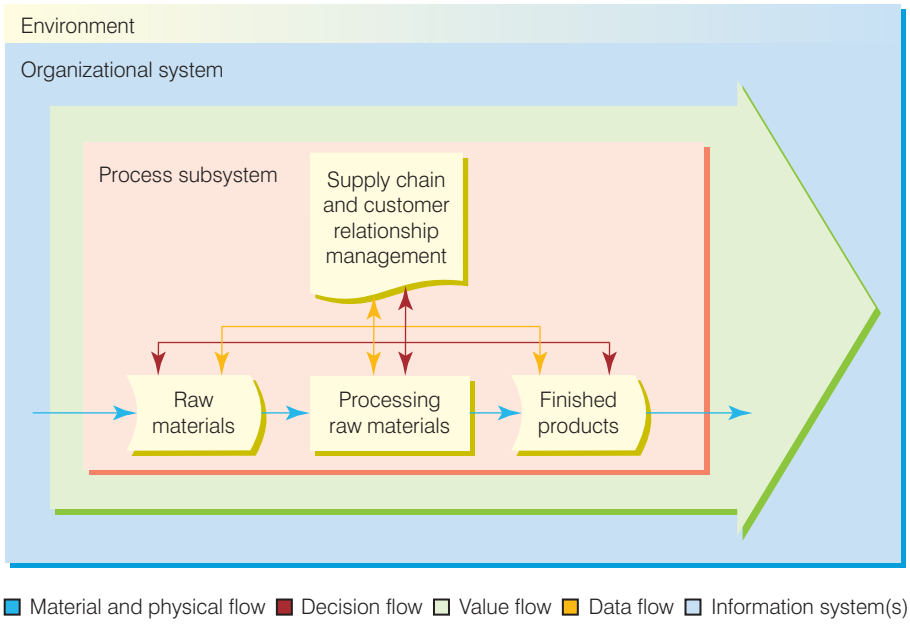


Figure 2.1 The Value Chain of a Manufacturing Company *Managing raw materials, inbound logistics, and warehouse and storage facilities is called 'upstream management', and managing finished product storage, outbound logistics, marketing and sales, and customer service is called 'downstream management'.*

The value chain is just as important (although it can be a little more difficult to apply) to companies that don't manufacture products but provide services, such as tax preparers and legal firms. By adding a significant amount of value to their products and services, companies ensure success.

Supply chain management (SCM) and customer relationship management (CRM) are two key parts of managing the value chain. SCM helps determine what supplies are required for the value chain, what quantities are needed to meet customer demand, how the supplies should be processed (manufactured) into finished goods and services, and how the shipment of supplies and products to customers should be scheduled, monitored and controlled.[3] For example, in the car manufacturing company mentioned previously, SCM can identify key suppliers and parts, negotiate with vendors for the best prices and support, make sure that all supplies and parts are available to manufacture cars and send finished products to dealerships around the country when they are needed. Increasingly, SCM is accomplished using the Internet and electronic marketplaces (e-marketplaces).[4] When an organization has many suppliers, it can use business-to-business exchanges. Zoodel, for example, facilitates international trade in the Middle East and elsewhere including Kazakhstan, Lebanon, Iran, China, Afghanistan, Iraq, Oman, Turkey and Russia. Their web portal can help buyers find the products they are looking for and facilitate communication between buyer and seller to allow them to negotiate, place orders and arrange delivery. Zoodel checks all sellers through a verification and inspection process and guarantees payment using an escrow system.[5]

CRM programs help a company manage all aspects of customer encounters, including marketing and advertising, sales, customer service after the sale and help to retain loyal customers. CRM can assist a company with collecting data on customers, contacting customers, informing them about new products and actively selling products to existing and new customers. Often, CRM software uses a variety of information sources, including sales from retail stores, surveys, email and Internet browsing habits, to compile comprehensive customer profiles. CRM systems can also collect customer feedback which can be used to design new products and services. Tesco, the UK's largest retail operation,[6] encourages its customers to use its Clubcard, which allows it to collect information on customer transactions. It uses this information to provide outstanding customer service and deliver loyalty rewards and perks to valued customers.[7] In return, customers are rewarded with discounts on Tesco products, holidays and other deals. Figure 2.2 shows the Tesco website which the company uses to help with CRM.

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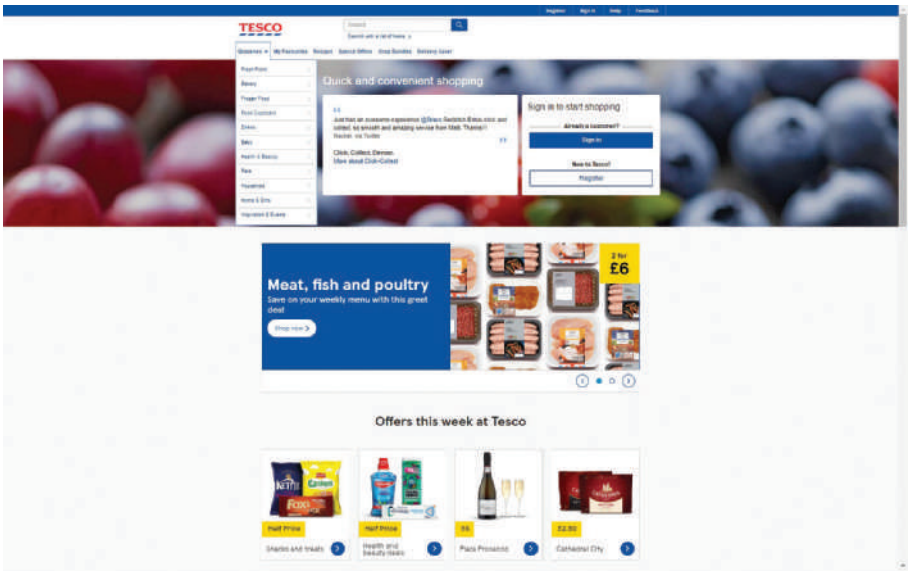


Figure 2.2 Tesco Website *Tesco uses its website to help with customer relationship management.*

What role does an information system play in these processes? A traditional view of information systems holds that organizations use them to control and monitor processes and ensure effectiveness and efficiency. Under this view, the output from a company's information systems is used to make changes to company processes. These changes could involve using different raw materials (inputs), designing new assembly-line procedures (product transformation) or developing new products and services (outputs). Here, the information system is external to the process and serves to monitor or control it.

A more contemporary view, however, holds that information systems are often so intimately involved that they are part of the process itself. From this perspective, the information system plays an integral role in the process, whether providing input, aiding product transformation or producing output. Consider a telephone directory business that creates telephone books for international businesses. A customer requests a telephone directory listing all steel suppliers in Western Europe. Using its information system, the directory business can sort files to find the suppliers' names and telephone numbers and organize them into an alphabetical list. The information system itself is an inseparable part of this process. It does not just monitor the process externally but works as part of the process to transform raw data into a product. In this example, the information system turns input (names and telephone numbers) into a sellable output (a telephone directory). The same system might also provide the input (the files storing the data) and output (printed pages for the directory).

This latter view provides a new perspective on how and why businesses can use information systems. Rather than attempting to understand information systems independently of the organization, we must consider the potential role of information systems within the process itself, often leading to the discovery of new and better ways to accomplish the process.

Organizational Structures

organizational structure
Organizational subunits and the way they relate to the overall organization.

Organizational structure refers to organizational subunits and the way they relate to each other. An organization's structure depends on its approach to management and can affect how it views and uses information systems. The types of organizational structures typically include traditional, project, team and virtual.

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Traditional Organizational Structure

A **traditional organizational structure**, also called a hierarchical structure, is like a managerial pyramid where the hierarchy of decision making and authority flows from the strategic management at the top, down to operational management and non-management employees. Compared to lower levels, the strategic level, including the managing director of the company and directors, has a higher degree of decision authority, more impact on business goals and more unique problems to solve (see Figure 2.3).

traditional organizational structure An organizational structure similar to a managerial pyramid, where the hierarchy of decision making and authority flows from strategic management at the top, down to operational management and non-management employees. Also called a hierarchical structure.

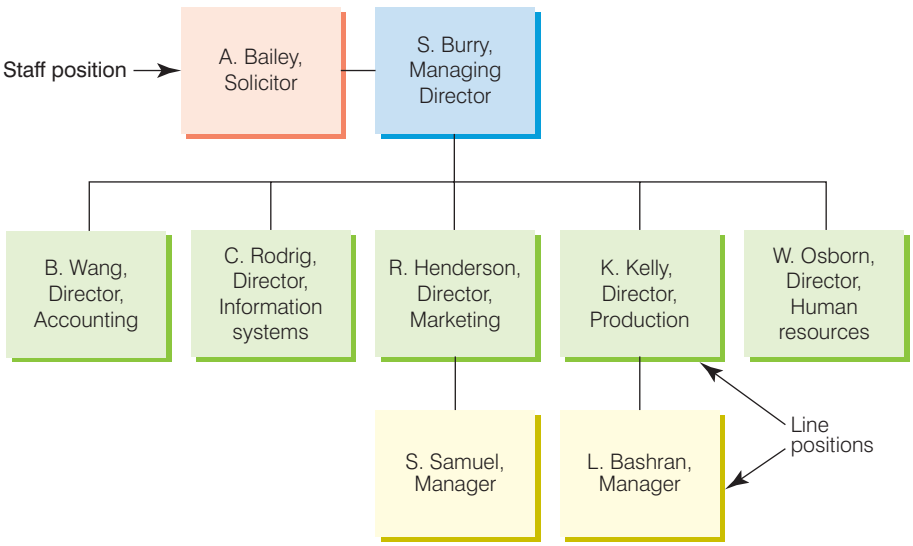


Figure 2.3 A Traditional Organizational Structure

In most cases, department heads report to a managing director or top-level manager. The departments are usually divided according to function and can include marketing, production, information systems, finance and accounting, research and development, and so on. The positions or departments that are directly associated with making, packing, or shipping goods are called line positions. A production manager who reports to a director of production is an example of a line position. Other positions might not be directly involved with the formal chain of command but instead assist a department or area. These are staff positions, such as a solicitor, reporting to the managing director.

Today, the trend is to reduce the number of management levels, or layers, in the traditional organizational structure. This type of structure, often called a **flat organizational structure**, empowers employees at lower levels to make decisions and solve problems without needing permission from mid-level managers. **Empowerment** gives employees and their managers more responsibility and authority to make decisions, take action and have more control over their jobs. For example, an empowered shop assistant can respond to customer requests and problems without needing permission from a manager. In a factory, empowerment might mean that an assembly-line worker can stop production to correct a problem before the product is passed to the next station.

flat organizational structure An organizational structure with a reduced number of management layers.

empowerment Giving employees and their managers more responsibility and the authority to make decisions, take certain actions and have more control over their jobs.

Information systems can be a key element in empowering employees because they provide the information employees need to make decisions. The employees might also be empowered to develop or use their own personal information systems, such as a simple forecasting model or spreadsheet.

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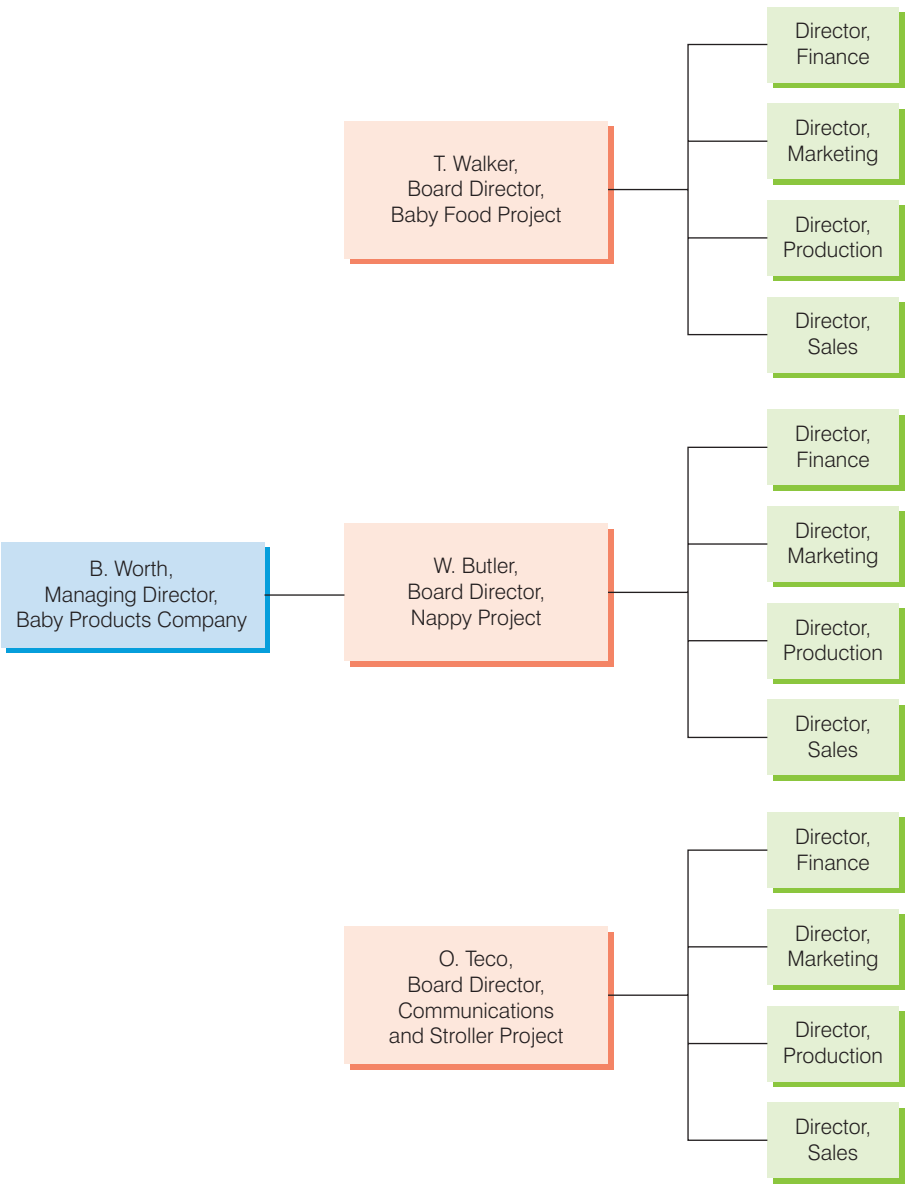
Project and Team Organizational Structures

project organizational structure A structure centred on major products or services.

A **project organizational structure** is centred on major products or services. For example, in a manufacturing firm that produces baby food and other baby products, each line is produced by a separate unit. Traditional functions such as marketing, finance and production are positioned within these major units (see Figure 2.4). Many project teams are temporary – when the project is complete, the members go on to new teams formed for another project.

2

Figure 2.4 A Project Organizational Structure



team organizational structure A structure centred on work teams or groups.

The **team organizational structure** is centred on work teams or groups. In some cases, these teams are small; in others, they are very large. Typically, each team has a leader who reports to an upper-level manager. Depending on its tasks, the team can be temporary or permanent. A healthcare company, for example, can form small teams to organize its administrators, physicians and others to work with individual patients.

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Virtual Organizational Structure

A **virtual organizational structure** is made up of individuals, teams or complete business units that work with other individuals, teams or complete business units in different geographic locations. This almost always requires the use of the Internet (or other telecommunications), and the teams can exist for a few weeks or years. The people involved might be in different countries and operating in different time zones. In other words, virtual organizational structures allow people who work together to be separated by location and time. The people might never meet physically, which explains the use of the word 'virtual'. In Chapter 10 we will examine some of the technologies that make this form of collaborative work possible. Despite their physical separation, members of a virtual organization can collaborate on any aspect of a project, such as supplying raw materials, producing goods and services, and delivering goods and services to the marketplace.

virtual organizational structure A structure that employs individuals, groups or complete business units in geographically dispersed areas that can last for a few weeks or years, often requiring telecommunications or the Internet.

A company can use a virtual organizational structure with its own dispersed workers who have distinct skills and abilities to reduce costs. PwC, a global accounting giant, uses virtual teams of 5 to 50 people in the learning and education department.[8] According to Peter Nicolas, the company's Learning Solutions manager, 'Virtual teaming is the norm for us'. The company takes advantage of software and technology, including Microsoft Live Meeting, Centra Software's Virtual-Classroom Application and Lotus Notes from IBM, to help the teams work from distant locations.

In addition to reducing costs or increasing revenues, a virtual organizational structure can provide an extra level of security. For instance, dispersing employees and using a virtual structure can provide an ability to deal with a disaster at the primary location. If this happened, the company would still have sufficient employees at other locations to keep the business running. Today's workers are performing company work at home, at a customer's location, in coffee shops, on pleasure boats and at convenient work centres in suburbia. People can work at any time. Using the Internet and email, workers can put the finishing touches to a new business proposal in Europe or Asia, while co-workers in North America are sleeping.

Successful virtual organizational structures share key characteristics. One strategy is to have in-house employees concentrate on the firm's core businesses and use virtual employees, groups or businesses to do everything else. Using information systems to manage the activities of a virtual structure is essential, often requiring specialized software to coordinate joint work. Even with sophisticated IS tools though, teams may still need face-to-face meetings, especially at the beginning of new projects.

Organizational Change

Most organizations are constantly undergoing change, both minor and major. The need for **organizational change** can be caused by internal factors, such as those initiated by employees at all levels, or external factors, such as activities wrought by competitors, stockholders, new laws, community regulations, natural occurrences (such as hurricanes) and general economic conditions. In the 1990s, the Internet caused massive changes in the way millions of organizations did business. The COVID-19 pandemic "lockdowns" and a rapid transition to homeworking in many countries throughout the world has changed work practices, certainly on a temporary level and possibly more permanently – an unforeseen environmental change.

organizational change The responses that are necessary so that for-profit and non-profit organizations can plan for, implement and handle change.

Change can be sustaining or disruptive. Sustaining change such as new or cheaper production equipment can help an organization improve its operations. For example, many factories are now able to use robots because prices for robots are falling and their useful lifetime is increasing, leading to a big market in second-hand robots.[9] Global Robots, for example, based in the UK, buys and sells industrial robots in large numbers from big manufacturers including ABB, Fanuc, Motoman and Kuka with the aim of making robots affordable to a wide market. They have agents in the USA, Scandinavia, Russia and Eastern Europe.[10]

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Disruptive change, on the other hand, often harms an organization's performance or even puts it out of business. In general, disruptive technologies might not originally have good performance, low cost or even strong demand. Over time, however, they often replace existing technologies. They can cause good, stable companies to fail when they don't change or adopt the new technology. Voice Over IP (VoIP) telephone technology is currently disrupting the business models of established companies such as BT (www.bt.com) who, in response, are moving towards providing broadband Internet connections as their main product.

Overcoming resistance to change, especially disruptive change, can be the hardest part of bringing information systems into a business. Occasionally, employees even attempt to sabotage a new information system because they do not want to learn the new procedures and commands. The best way to avoid this resistance is to involve the employees in the decision to implement the change, and consult them on the development or purchase of the information system.

When a company introduces a new information system, a few members of the organization must become agents of change – champions of the new system and its benefits. Understanding the dynamics of change can help them confront and overcome resistance so that the new system can be used to maximum efficiency and effectiveness.

A significant portion of an organization's expenses are used to hire, train and compensate talented staff. So organizations try to control costs by determining the number of employees they need to maintain high-quality goods and services. Strategies to contain costs are outsourcing, on-demand computing and downsizing.

outsourcing Contracting with outside professional services to meet specific business needs.

Outsourcing involves contracting with outside professional services to meet specific business needs. Often, companies outsource a specific business process, such as recruiting and hiring employees, developing advertising materials, promoting product sales or setting up a global telecommunications network. Organizations often outsource a process to focus more closely on their core business and target limited resources to meet strategic goals. South Africa has a sophisticated manufacturing and service-based economy, and many small businesses from the UK and Australia are outsourcing their services to that country. In addition to a highly skilled workforce, there are time zone advantages too (everyone is awake at the same time for at least part of the day), and potential cost savings. The Shopper Collective in Australia, which helps its clients to understand their customers better, outsources its data analytics to South Africa.[11, 12]

Other reasons for outsourcing are to trim expenses or benefit from the expertise of a service provider. A growing number of organizations, however, are finding that outsourcing does not necessarily lead to reduced costs. One of the primary reasons for cost increases is poorly written contracts that tack on charges from the outsourcing vendor for each additional task. Other potential drawbacks of outsourcing include loss of control and flexibility, overlooked opportunities to strengthen core competency and low employee morale.

on-demand computing Contracting for computer resources to rapidly respond to an organization's varying workflow. Also called on-demand business and utility computing.

On-demand computing is an extension of the outsourcing approach, and many companies offer on-demand computing to business clients and customers. On-demand computing, also called on-demand business and utility computing, involves rapidly responding to the organization's flow of work as the need for computer resources varies. It is often called 'utility computing' because the organization pays for computing resources from a computer or consulting company just as it pays for electricity from a utility company. This approach treats the information system – including hardware, software, databases, telecommunications, personnel and other components – more as a service than as separate products. In other words, instead of purchasing hardware, software and database systems, the organization only pays a fee for the systems it needs at peak times. The approach can save money because the organization does not pay for systems that it doesn't routinely need. It also allows the organization's IS staff to concentrate on more strategic issues.

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Downsizing involves reducing the number of employees to cut costs. The term 'rightsizing' is also used. Rather than pick a specific business process to downsize, companies usually look to downsize across the entire company. Downsizing clearly reduces total payroll costs, although employee morale can suffer.

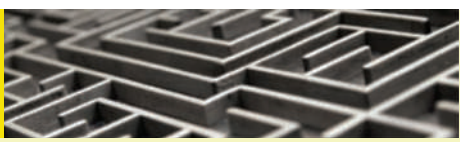
downsizing Reducing the number of employees to cut costs.

Employers need to be open to alternatives to reducing the number of employees and use layoffs as the last resort. It's simpler to encourage people to leave voluntarily through early retirement or other incentives. Voluntary downsizing programmes often include a buyout package offered to certain classes of employees (for example, those over 50 years old). The buyout package offers employees certain benefits and cash incentives if they voluntarily retire from the company. Other options are job sharing and transfers.

Organizational learning is closely related to organizational change. According to the concept of organizational learning, organizations adapt to new conditions or alter their practices over time. Assembly-line workers, secretaries, shop assistants, managers and executives all learn better ways of doing business and incorporate them into their day-to-day activities. Collectively, these adjustments based on experience and ideas are called 'organizational learning'. In some cases, the adjustments can be a radical redesign of business processes, often called 'reengineering'. In other cases, these adjustments can be more incremental, a concept called 'continuous improvement'. Both adjustments reflect an organization's strategy: the long-term plan of action for achieving their goals.

organizational learning The adaptations to new conditions or alterations of organizational practices over time.

Ethical and Societal Issues



Customers Out of Pocket After Direct Transfer Error

A Direct Debit is an instruction from a customer to their bank giving permission for a third party to collect varying amounts from their account, usually in order to pay regular bills. In the UK, the Direct Debit Guarantee applies to all banks and building societies and protects customers in the event of errors made in payments such as an incorrect amount being taken or a payment being made on the wrong date. The guarantee says that, 'If an error is made in the payment of your Direct Debit, by the organisation or your bank or building society, you are entitled to a full and immediate refund of the amount paid from your bank or building society'. Many Internet banking customers assume that the guarantee applies to online transfers that they have made, but in fact this is not the case.

At the end of 2014, a record label manager and her husband fell victim to a sophisticated email scam. The couple had some work done to their house and received an email from what they thought was their builder giving them instructions on how to pay. The email asked for over £25,000 to be transferred and gave the account number and sort code of the bank it was to be paid into. Later that day, they arranged payment and thought no more of it, until they received a second email chasing payment. At that point they looked into it and realized the first email had been sent by a scammer. The email address had been slightly different: 'development' instead of 'developments' in the company name.

The money had been sent from Lloyds Bank to Barclays Bank. Contacting Lloyds, they were told that the bank had simply followed their instructions. So they turned to Barclays. Initially, the bank refused to give them any information, citing that they had to protect the identity of the fraudster in line with data protection legislation. They did learn something though. 'Our own questioning with the banks had traced our money to an account in the north [of England] owned by a man called "Harry". It, of course, had been immediately fully withdrawn and closed', said Sarah Fisher, the victim. Eventually, Barclays said that by the time it was alerted, the couple's £25,000 had been 'utilized' by the account holder, so it was unable to return any of their cash.

(*continued*)

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Its letter added: 'We do not report scam claims to the police because the bank is not the victim'. However, they also said they would cooperate fully with the police as part of a criminal investigation.

Following up on this story, *The Guardian* newspaper sent one of their journalists to open an account at Barclays. 'It was impossible without giving the bank my full name, date of birth, addresses for the past three years, email address, phone number, income and employment details, including my full company address. It also required permission from me to make checks on everything I submitted'.

When the couple reported the incident to the police, they did not pursue an investigation but instead referred the case to Action Fraud, the UK's national fraud and cybercrime reporting centre. 'The reason the police are citing for not pursuing an investigation is that this type of crime is so rife they haven't got the resources – they say that compared to some of the cases they are dealing with, it's small scale', says Mrs Fisher. City of London Police Commander Chris Greany concedes that this type of 'invoicing fraud', where people are fooled into transferring funds, is an emerging threat. 'Those with viable lines of inquiry have been sent to local police forces to investigate', he says. The name for this kind of fraud is an Authorised Push Payment Scam, and has become common. The story above has been repeated many times, eventually causing UK banks to act. In February 2018, the Payment Systems Regulator created a steering committee to develop a code of practice to guide how the financial sector should respond to reported cases. In May 2019 the new code of practice was published, recommending that victims should be reimbursed unless they ignored their bank's warnings or were grossly negligent in transferring the money. While signing up to the code is voluntary, many of the main high street players joined immediately. If one of their customers is scammed, the bank will have to decide whether or not to refund them within 15 working days, and then make the refund as soon as possible. Banks should also be proactive, identifying customers whom they feel are at a high risk of being scammed and warning them in advance. They should also delay payments while they investigate potential scams, and freeze accounts they believe might belong to scammers.

All of this comes too late to help Sarah Fisher though.

Questions

1. How could banks identify customers who are likely to be victims?
2. How could delayed payments be a problem for banks?
3. What additional controls could banks offer their customers?
4. How should you assess emails that ask for money transfers?

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Reengineering and Continuous Improvement

To stay competitive, organizations must occasionally make fundamental changes to the way they do business. In other words, they must change the activities, tasks or processes they use to achieve their goals. **Reengineering**, also called 'process redesign' and 'business process reengineering' (BPR), involves the radical redesign of business processes, organizational structures, information systems and values of the organization to achieve a breakthrough in business results. Reengineering can reduce delivery times, increase product and service quality, enhance customer satisfaction, and increase revenues and profitability. When the Mittal Steel plant in Vanderbijlpark, South Africa, reengineered its steelmaking operations, introducing new automated systems, it was able to reduce the amount of raw materials used in its processes and increase its output of steel, which was of a higher quality than before. It was also able to reduce processing time and improve plant availability.[13]

reengineering Also known as 'process redesign' and 'business process reengineering' (BPR). The radical redesign of business processes, organizational structures, information systems and values of the organization to achieve a breakthrough in business results.

In contrast to simply automating the existing work process, reengineering challenges the fundamental assumptions governing its design. It requires finding and vigorously challenging old rules blocking major business process changes. These rules are like anchors weighing down a firm and keeping it from competing effectively. Table 2.1 provides some examples of such rules.

Table 2.1 Selected Business Rules that Affect Business Processes

Rule	Original Rationale	Potential Problem
Hold small orders until full lorry load shipments can be assembled	Reduce delivery costs	Customer delivery is slow
Do not accept an order until customer credit is approved	Reduce potential for bad debt	Customer service is poor
Let headquarters make all merchandizing decisions	Reduce number of items carried in inventory	Customers perceive organization has limited product selection

In contrast to reengineering, the idea of **continuous improvement** is to constantly seek ways to improve business processes and add value to products and services. This continual change will increase customer satisfaction and loyalty and ensure long-term profitability. Manufacturing companies make continual product changes and improvements. Service organizations regularly find ways to provide faster and more effective assistance to customers. By doing so, these companies increase customer loyalty, minimize the chance of customer dissatisfaction and diminish the opportunity for competitive inroads. Table 2.2 compares these two strategies.

continuous improvement Constantly seeking ways to improve business processes to add value to products and services.

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Table 2.2 Comparing Business Process Reengineering and Continuous Improvement

Business Process Reengineering	Continuous Improvement
Strong action taken to solve serious problems	Routine action taken to make minor improvements
Top-down change driven by senior executives	Bottom-up change driven by workers
Broad in scope; cuts across departments	Narrow in scope; focus is on tasks in a given area
Goal is to achieve a major breakthrough	Goal is continuous, gradual improvements
Often led by outsiders	Usually led by workers close to the business
Information system integral to the solution	Information systems provide data to guide the improvement team

User Satisfaction and Technology Acceptance

To be effective, reengineering and continuous improvement efforts must result in satisfied users and be accepted and used throughout the organization. You can determine the actual usage of an information system by the amount of technology diffusion and infusion.

technology diffusion A measure of how widely technology is spread throughout the organization.

Technology diffusion is a measure of how widely technology is spread throughout an organization. An organization has a high level of technology diffusion if computers and information systems are located in most departments. Some online merchants, such as BT (www.bt.com), have a high diffusion and use computer systems to perform most of their business functions, including marketing, purchasing and billing.

technology infusion The extent to which technology is deeply integrated into an area or department.

Technology infusion, on the other hand, is the extent to which technology permeates an area or department. In other words, it is a measure of how deeply embedded technology is in an area of the organization. Some architectural firms, for example, use computers in all aspects of designing a building from drafting to final blueprints. The design area, thus, has a high level of infusion. Of course, a firm can have a high level of infusion in one part of its operations and a low level of diffusion overall. The architectural firm might use computers in all aspects of design (high infusion in the design area), but not to perform other business functions, including billing, purchasing and marketing (low diffusion). Diffusion and infusion often depend on the technology available now and in the future, the size and type of the organization, and the environmental factors that include the competition, government regulations, suppliers and so on. This is often called the 'technology, organization and environment' (TOE) framework.[14]

An active research area in IS involves identifying why people accept and use one system but dislike and therefore don't use another. One early model, the Technology Acceptance Model (TAM), shows that people will use a system if it is easy to use and useful to them. This in itself is unhelpful to IS developers; however, TAM has been the basis for a large body of research that is ongoing and which hopes to produce more practical results.

Although an organization might have a high level of diffusion and infusion with computers throughout the organization, this does not necessarily mean that information systems are being used to their full potential.

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Information Systems @ Work



Neto Helps Australian Small Businesses Get Online

In 2011, Australian entrepreneur Ryan Murtagh saw an opportunity to help get small businesses online. Having been an e-retailer, he realized the struggle involved in scaling up an e-commerce store efficiently. This set him on the path to developing an end-to-end e-commerce platform designed to automate and streamline online trading for small to medium sized businesses. By focusing on the 'pain points' he had experienced himself, he assembled a team and set about solving these problems through software, using a company called Neto. Five years later, thousands of businesses use their platform to run online stores, and cumulatively they have turned over more than €1 billion through the service. One of these companies is Bicycles Online.

Founded by Jonathon Allara and James van Rooyen, Bicycles Online is an Australian owned, online speciality retailer of road bikes, mountain bikes, bike parts, accessories and clothing. As the business began to grow, the two were faced with a number of business obstacles. They sold their products through multiple channels: their own website, eBay, a number of shopping comparison websites and industry specific marketplaces. They also had just expanded from one to two warehouses. They needed an e-commerce solution that allowed them to accurately and effectively manage stock and fulfil orders across multiple warehouse locations and multiple sales channels, preferably one that integrated well with freight providers and Australia Post. Searching online they stumbled across Neto.

'With Neto, all integrations that we need are built into the system and we also have the ability to build custom integrations with Neto to other marketplaces if need be. The in-built shipping labelling and manifesting as well as the real-time shipping quotes and multi-carrier support have had a significant impact on our business. The time taken for us to fulfil orders has halved and we have also experienced significant cost savings with our freight charges by being able to have the option to choose the most cost effective freight option on an order by order basis', said Mr Allara.

Bicycles Online prides itself on offering the same level of services that customers would get in a brick-and-mortar store, or better, combined with competitive pricing. These services include free shipping, 14-day test rides and a national network of centres where bikes can be serviced. By using Neto, Bicycles Online gets to focus on what it knows – bikes – and uses Neto to focus on what it knows – online platforms.



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Using Neto, netohq.com, they were able to create and customize what Neto refers to as their 'e-commerce ecosystem', which means their entire online presence. Using a combination of eBay, accounting software Xero, Australia Post and help desk software Zendesk, they were able to streamline their business processes and increase sales.

'By moving over our platform to Neto we have been able to seamlessly manage our order fulfilment and inventory management across two warehouses', said Mr Allara. 'Neto has provided us with a solution that centralises our sales channel management – we can sell bikes online, whether it is on our main website, eBay or other marketplaces. When we were operating on our previous e-commerce system, integration with eBay and other marketplaces was very difficult and required linking up with third-party systems to try and do so – which can be costly and sometimes unreliable.'

Australia's largest telecommunications and media company Telstra Corporation Limited has taken notice and recently invested in Neto. Speaking in 2015, Telstra Business Group Managing Director, Will Irving, said, 'We estimate that by 2020, 82 per cent of SMEs will have a web presence, rising from 58 per cent today. The role of e-commerce will only become more important in the competitive retail environment. The Neto platform allows businesses to meet the needs of customers at their convenience: where they want and when they want'. Mr Murtagh said, 'With the technical expertise and scale that Telstra brings, I believe that this partnership will allow us to accelerate our development plans and to provide Australian SMEs with an even better product and a higher level of service and support'.

Questions

1. Why is it important for Bicycles Online to integrate with freight providers?
2. How can a company's social media presence integrate well with its retail website?
3. Why would Telstra be interested in Neto?
4. Why would Bicycles Online sell through so many channels?

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The Applications Portfolio

applications portfolio A scheme for classifying information systems according to the contribution they make to the organization.

In Chapter 1 we looked at how information systems can be classified by the management level of the user. The **applications portfolio** is perhaps a more useful classification scheme. It sorts information systems according to the contribution they make to the business. According to the applications portfolio, there are four types of system:

support applications Support applications make work more convenient but are not essential.

1. Support: **Support applications** are nice to have, but are not essential. They include things that are convenient, but without them the organization can still conduct business. Typical support applications include electronic diaries and instant messaging software, used to let employees in an office communicate with each other.

key operational applications Key operational applications are essential. Without them the organization could not conduct business.

2. Key operational: **Key operational applications** are essential. Without them the organization would not be able to do business. Transaction processing systems, mentioned in Chapter 1 and discussed fully in Chapter 8, are an example. If the checkout system at a Tesco shop malfunctions, Tesco would be unable to sell goods until it was repaired. The website of every e-commerce business is key operational.

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3 Strategic: A **strategic application** is an information system that gives a business an advantage over some or all of its competitors. Some ideas for what this advantage might be are discussed later in this chapter in the section on Competitive Advantage. The term 'strategic' should not be confused with the same term used to describe senior management in a business. A strategic system could appear anywhere in the company hierarchy.

strategic application A strategic application gives a firm a competitive advantage.

4 Future strategic: A **future strategic application** (also known as a 'potential strategic' or 'high potential' application) is an idea for, or a prototype of, an information system which, if developed, might one day become a strategic system. A company may have ten future strategic systems and decide to invest in only one. This decision is often a judgement call made by senior management. It may be that the technology to develop a future strategic system is currently too expensive and the company is waiting for prices to fall.

future strategic application Future strategic applications are ideas for systems which, if fully developed and deployed, might one day become strategic applications.

There is an endless cycle at work with systems starting life in one part of the portfolio and finishing in another. Typically an innovative, leading company will come up with an idea for a potential strategic system. If they invest in it and it is successful, it becomes a strategic system. Their competitors see that they have an advantage and so create their own versions of the system. Eventually, the system will become industry standard and then be key operational. In the meantime, the innovative company will have had more ideas for future strategic systems, and so the cycle starts again. Companies that see themselves as industry followers rather than industry leaders will not have strategic or future strategic systems in their portfolio.

Success Factors

Many writers have suggested reasons why some information systems are implemented successfully and others are not. It is of vital importance that a company's information systems are aligned with the company's goals. Misalignment is a frequently cited reason for information systems failure. The main way of achieving **alignment** is for senior managers to consider the business processes they have in place to achieve company goals, and ask what information systems are needed to support these business processes. Less frequently, a business, typically a small business or even a single entrepreneur, will consider what technology is available and ask what business goals can be achieved using it. In this case, information technology is dictating business strategy instead of business strategy dictating what information technology is used. Both are valid paths to alignment.

alignment When the output from an information system is exactly what is needed to help a company achieve its strategic goals, the two are said to be in alignment.

Other common success factors are:

- Senior management must be committed to the development or purchase of the information system and support it fully.
- End-users of the system should be involved as early and as much as possible in the development or purchase of the system.
- Time must be taken to carefully determine what the system must do, something known as **requirements engineering**. Requirements must be clearly stated and understood and accepted by everyone involved.
- Strong project management in the development or purchase of the information system.

requirements engineering Also known as 'requirements analysis' and 'requirements capture'. Identifying what an information system is needed (required) to do. Once the requirements have been identified, a solution can then be designed.

Later on in this text we will examine Joint Application Development, a method for creating IS which places users at the centre of the development.

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2.2 Competitive Advantage

competitive advantage The ability of a firm to outperform its industry; that is, to earn a higher rate of profit than the industry norm.

A **competitive advantage** is the ability of a firm to outperform its industry, that is, to earn a higher rate of profit than the industry norm[15] and can result from higher-quality products, better customer service and lower costs. Establishing and maintaining a competitive advantage is complex. An organization often uses its information system to help it do this. Ultimately, it is not how much a company spends on information systems but how it makes and manages investments in technology. Companies can spend less and get more value.

Factors That Lead Firms to Seek Competitive Advantage

A number of factors can lead a company to seek to attain a competitive advantage. Michael Porter, a prominent management theorist, suggested a simple but widely accepted model of the competitive forces in an industry, also called the **five-forces model**. A strong force can put a business at a disadvantage and lead it to invest in technology that can weaken it. The five forces are: (1) the rivalry among existing competitors, (2) the threat of new entrants, (3) the threat of substitute products and services, (4) the bargaining power of buyers, and (5) the bargaining power of suppliers. The more these forces combine in any instance, the more likely firms will seek competitive advantage and the more dramatic the results of such an advantage will be.

five-forces model A widely accepted model that identifies five key factors that can lead to attainment of competitive advantage, including (1) the rivalry among existing competitors, (2) the threat of new entrants, (3) the threat of substitute products and services, (4) the bargaining power of buyers, and (5) the bargaining power of suppliers.

Given the five market forces just mentioned, Porter and others have proposed a number of strategies to attain competitive advantage, including cost leadership, differentiation, niche strategy, altering the industry structure, creating new products and services, and improving existing product lines and services.[16] In some cases, one of these strategies becomes dominant. For example, with a cost leadership strategy, cost can be the key consideration at the expense of other factors if needs be.

Cost Leadership

The intent of a cost leadership strategy is to deliver the lowest possible products and services cost. In the UK, supermarket Asda has used this strategy for years. Cost leadership is often achieved by reducing the costs of raw materials through aggressive negotiations with suppliers, becoming more efficient with production and manufacturing processes, and reducing warehousing and shipping costs. Some companies use outsourcing to cut costs when making products or completing services.

Differentiation

The intent of differentiation as a strategy is to deliver different products and services. This strategy can involve producing a variety of products, giving customers more choice, or delivering higher-quality products and services. Many car companies make different models that use the same basic parts and components, giving customers more options. Other car companies attempt to increase perceived quality and safety to differentiate their products. Some consumers are willing to pay higher prices for vehicles that differentiate on higher quality or better safety.

Niche Strategy

A niche strategy will deliver to only a small, niche market. Porsche, for example, doesn't produce inexpensive estate cars or saloons. It makes high-performance sports cars and four-wheel drives. Rolex only makes high-quality, expensive watches. It doesn't make inexpensive, plastic watches that can be purchased for €20 or less.

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Altering the Industry Structure

Changing the industry to become more favourable to the company or organization is another strategy companies use. The introduction of low-fare airline carriers, such as easyJet, has forever changed the airline industry, making it difficult for traditional airlines to make high profit margins. To fight back, airlines such as British Airways cut their flight prices and started to emphasize their strengths over low-cost airlines in their advertising. These include landing in central airports rather than airports many miles out of the city they supposedly serve, and extra staff and resources to cope if there is a fault with an aircraft, or adverse weather grounds all planes. Creating **strategic alliances** can also alter the industry structure. A strategic alliance, also called a 'strategic partnership', is an agreement between two or more companies that involves the joint production and distribution of goods and services.

strategic alliances (strategic partnership) An agreement between two or more companies that involves the joint production and distribution of goods and services.

Creating New Products and Services

Some companies introduce new products and services periodically or frequently as part of their strategy. This strategy can help a firm gain a competitive advantage, especially in the computer industry and other high-tech businesses. If an organization does not introduce new products and services every few months, the company can quickly stagnate, lose market share and decline. Companies that stay on top are constantly developing new products and services.

Improving Existing Product Lines and Services

Making real or perceived improvements to existing product lines and services is another strategy. Manufacturers of household products are always advertising 'new and improved' products. In some cases, the improvements are more perceived than real refinements; usually, only minor changes are made to the existing product, such as reducing the amount of sugar in a breakfast cereal. Some mail order companies are improving their service by using Radio Frequency Identification (RFID) tags to identify and track the location of their products as they are shipped from one location to another. Customers and managers can instantly locate products as they are shipped from suppliers to the company, to warehouses and finally to customers.

Other potentially successful strategies include being the first to market, offering customized products and services, and hiring talented staff, the assumption being that the best people will determine the best products and services to deliver to the market and the best approach to deliver these products and services. Companies can also combine one or more of these strategies.

2.3 Evaluating IS

Once an information system has been implemented, management will want to assess how successful it has been in achieving its goals. Often this is a difficult thing to do, and many businesses do not attempt to take anything more than an informal approach to evaluation.[17] Businesses can use measurements of productivity, return on investment (ROI), net present value and other measures of performance to evaluate the contributions their information systems make to their businesses.

Productivity

Developing information systems that measure and control productivity is a key element for most organizations. **Productivity** is a measure of the output achieved divided by the input required. A higher level of output for a given level of input means greater productivity; a lower level of output for a given level of input means lower productivity. The numbers assigned to productivity levels are not

productivity A measure of the output achieved divided by the input required. Productivity = (Output ÷ Input) × 100%.

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always based on labour hours – productivity can be based on factors such as the amount of raw materials used, resulting quality, or time to produce the goods or service. The value of the productivity number is not as significant as how it compares with other time periods, settings and organizations.

After a basic level of productivity is measured, an information system can monitor and compare it over time to see whether productivity is increasing. Then a company can take corrective action if productivity drops below certain levels. In addition to measuring productivity, an information system can be used within a process to significantly increase productivity. Thus, improved productivity can result in faster customer response, lower costs and increased customer satisfaction.

2

In the late 1980s and early 1990s, overall productivity did not seem to improve as a company increased its investments in information systems. Often called the productivity paradox, this situation troubled many economists who were expecting to see dramatic productivity gains. In the early 2000s, however, productivity again seemed to be on the rise.

Return on Investment and the Value of Information Systems

return on investment (ROI)
One measure of IS value that investigates the additional profits or benefits that are generated as a percentage of the investment in IS technology.

One measure of IS value is **return on investment (ROI)**. This measure investigates the additional profits or benefits that are generated as a percentage of the investment in IS technology. A small business that generates an additional profit of €20,000 for the year as a result of an investment of €100,000 for additional computer equipment and software would have a return on investment of 20 per cent (€20,000/€100,000). In many cases, however, it can be difficult to accurately measure ROI.[18]

Earnings Growth

Another measure of IS value is the increase in profit or earnings growth it brings. For instance, a mail-order company might install an order-processing system that generates a 7 per cent earnings growth compared with the previous year.

Market Share

Market share is the percentage of sales that a product or service has in relation to the total market. If installing a new online catalogue increases sales, it might help a company to increase its market share by 20 per cent.

Customer Awareness and Satisfaction

Although customer satisfaction can be difficult to quantify, about half of today's best global companies measure the performance of their information systems based on feedback from internal and external users. Some companies use surveys and questionnaires to determine whether the IS investment has increased customer awareness and satisfaction.

Total Cost of Ownership

total cost of ownership (TCO)
The measurement of the total cost of owning computer equipment, including desktop computers, networks and large computers.

Another way to measure the value of information systems was developed by the Gartner Group and is called the **total cost of ownership (TCO)**. This approach breaks down total costs into areas such as the cost to acquire the technology, technical support, administrative costs and end-user operations. Other costs in TCO include retooling and training costs. TCO can help to develop a more accurate estimate of the total costs for systems that range from desktop computers to large mainframe systems. Market research groups often use TCO to compare products and services.

ROI, earnings growth, market share, customer satisfaction and TCO are only a few measures that companies use to plan for and maximize the value of their IS investments. Regardless of the

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difficulties, organizations must attempt to evaluate the contributions that information systems make to assess their progress and plan for the future. Information technology and personnel are too important to leave to chance.

Risk

In addition to the ROI measures of a new or modified information system, managers should also consider the risks of designing, developing and implementing these systems. Information systems can sometimes be costly failures. Some companies, for example, have attempted to implement enterprise resource planning (ERP) systems (see Chapter 7) and failed, costing them millions of euros. In other cases, e-commerce applications have been implemented with little success. The costs of development and implementation can be greater than the returns from the new system.

2.4 Careers in Information Systems

Realizing the benefits of any information system requires competent and motivated IS personnel, and many companies offer excellent job opportunities. Professionals with careers in information systems typically work in an IS department as web developers, computer programmers, systems analysts, database developers and administrators, computer operators, technical support or in other positions. In addition to technical skills, they need skills in written and verbal communication, an understanding of organizations and the way they operate, and the ability to work with people and in groups. Today, many good information, business and computer science schools require these business and communications skills of their graduates.

It is not uncommon for IS personnel to undertake remote working, where colleagues are not co-located but are in separated geographical areas, using many of the information technologies described in this book to facilitate this. Contract work, where IS professionals complete short-term contracts, maybe for six months, are also not unusual.

In general, IS professionals are charged with maintaining the broadest perspective on organizational goals. Most medium to large organizations manage information resources through an IS department. In smaller businesses, one or more people might manage information resources, with support from outsourced services. As shown in Figure 2.5, the IS department has three primary responsibilities: operations, systems development and support.

Operations

People in the operations component of a typical IS department work with information systems in corporate or business unit computer facilities. They tend to focus more on the efficiency of IS functions rather than their effectiveness.

System operators primarily run and maintain IS equipment and are typically trained at technical schools or through on-the-job experience. They are responsible for starting, stopping and correctly operating mainframe systems, networks, back-up drives, disc devices, printers and so on. Other operations include scheduling, hardware maintenance and preparing input and output. Data-entry operators convert data into a form the computer system can use. They can use terminals or other devices to enter business transactions, such as sales orders and payroll data. Increasingly, data entry is being automated – captured at the source of the transaction rather than entered later. In addition, companies might have local area network (LAN) and web operators who run the local network and any websites the company has.

Systems Development

The systems development component of a typical IS department focuses on specific development projects and ongoing maintenance and review. Systems analysts and

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programmers, for example, address these concerns to achieve and maintain IS effectiveness. The role of a systems analyst is multifaceted. Systems analysts help users determine what outputs they need from the system and construct plans for developing the necessary programs that produce these outputs. Systems analysts then work with one or more programmers to make sure that the appropriate programs are purchased, modified from existing programs or developed. A computer programmer uses the plans the systems analyst has created to develop or adapt one or more computer programs that produce the desired outputs. Often, analysts will be using an agile approach to software development, where software is written rapidly and then continuously improved. One such approach is called Scrum where small teams intensively work together on small tasks that can be completed within one month.

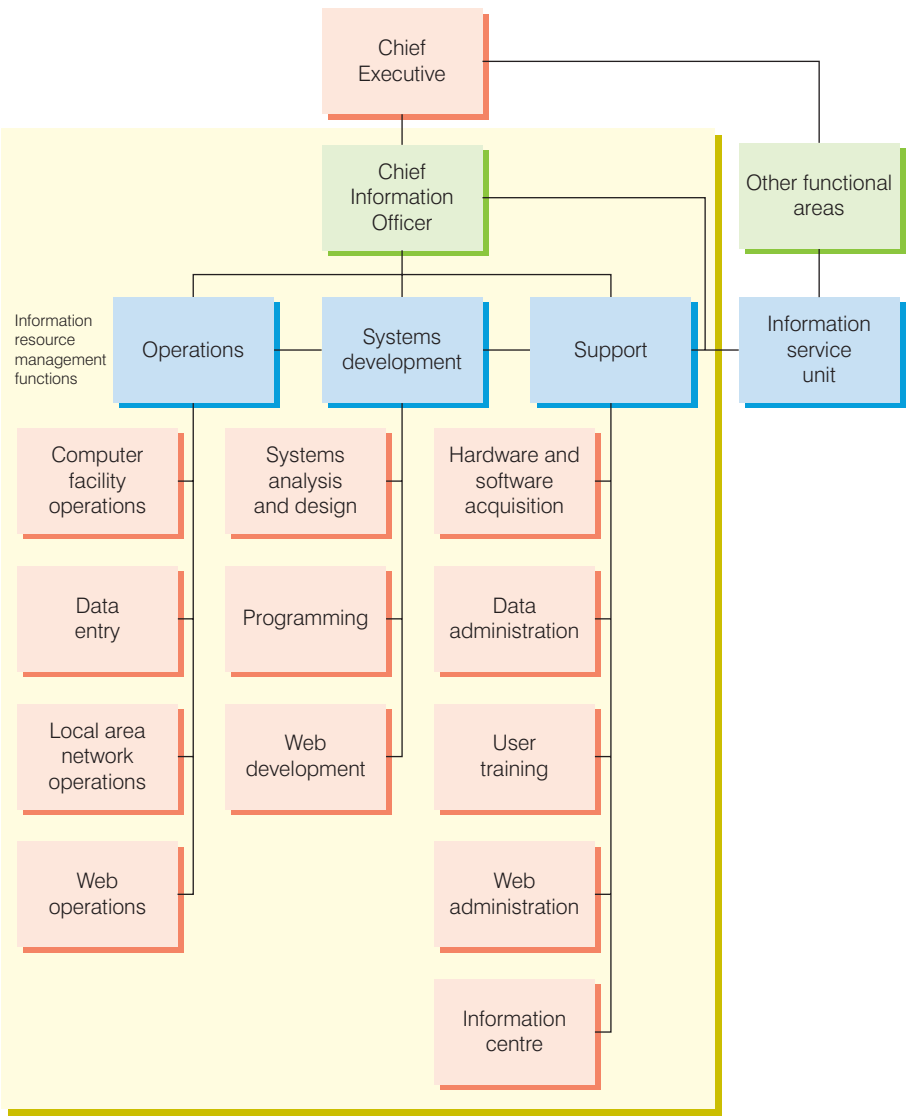


Figure 2.5 The IS Department *Not all businesses will have all of these divisions, and in many firms people will take on multiple roles.*

With the dramatic increase in the use of the Internet, intranets and extranets, many companies have web or Internet developers who create effective and attractive websites for customers, internal personnel, suppliers, stockholders and others who have a business relationship with the company.

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Support

The support component of a typical IS department provides user assistance in hardware and software acquisition and use, data administration, user training and assistance, and web administration. In many cases, support is delivered through an information centre.

Because IS hardware and software are costly, a specialized support group often manages computer hardware and software acquisitions. This group sets guidelines and standards for the rest of the organization to follow in making purchases. It must gain and maintain an understanding of available technology and develop good relationships with vendors.

A database administrator focuses on planning, policies and procedures regarding the use of corporate data and information. For example, database administrators develop and disseminate information about the organization's databases for developers of IS applications. In addition, the database administrator monitors and controls database use.

User training is key to getting the most from any information system, and the support area ensures that appropriate training is available. Training can be provided by internal staff or from external sources. For example, internal support staff can train managers and employees in the best way to enter sales orders, to receive computerized inventory reports and to submit expense reports electronically. Companies also hire outside firms to help train users in other areas, including the use of word processing, spreadsheets and database programs.

Web administration is another key area for support staff. With the increased use of the Internet, web administrators are sometimes asked to regulate and monitor Internet use by employees and managers to make sure that it is authorized and appropriate. Web administrators also maintain the organization's website to keep it accurate and current, which can require substantial resources.

The support component typically operates the helpdesk. A helpdesk provides users with assistance, training, application development, documentation, equipment selection and setup, standards, technical assistance and troubleshooting.

Information Service Units

An information service unit is basically a miniature IS department attached and directly reporting to a functional area in a large organization. Notice the information service unit shown in Figure 2.5. Even though this unit is usually staffed by IS professionals, the project assignments and the resources necessary to accomplish these projects are provided by the functional area to which it reports. Depending on the policies of the organization, the salaries of IS professionals staffing the information service unit might be budgeted to either the IS department or the functional area.

Typical IS Titles and Functions

The organizational chart shown in Figure 2.5 is a simplified model of an IS department in a typical medium or large organization. Many organizations have even larger departments, with increasingly specialized positions such as librarian or quality assurance manager. Smaller firms often combine the roles shown in Figure 2.5 into fewer formal positions.

The Chief Information Officer

The role of the chief information officer (CIO) is to employ an IS department's equipment and personnel to help the organization attain its goals. The CIO is a senior manager concerned with the overall needs of the organization who sets organization-wide policies, and plans, manages and acquires information systems. Some of the CIO's top concerns include integrating IS operations with business strategies, keeping up with the rapid pace of technology, and defining and assessing the value of systems development projects. The high level of the CIO position

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reflects that information is one of the organization's most important resources. A CIO works with other high-level officers in an organization, including the finance director and the chief executive officer, in managing and controlling total corporate resources. CIOs must also work closely with advisory committees, stressing effectiveness and teamwork and viewing information systems as an integral part of the organization's business processes – not an adjunct to the organization. Thus, CIOs need both technical and business skills.

Administrators

local area network (LAN) A computer network that connects computer systems and devices within a small area, such as an office, home or several floors in a building.

Local area network (LAN) administrators set up and manage the network hardware, software and security processes. They manage the addition of new users, software and devices to the network. They also isolate and fix operational problems. LAN administrators are in high demand and often solve both technical and non-technical problems. Database administrators manage the use, maintenance and security of a company's databases. Often a database administrator will help users extract the data they need in the format they require.

Internet Careers

These careers are in the areas of web operations, web development and web administration. As with other areas in IS, many top-level administrative jobs are related to the Internet. These career opportunities are found in both traditional companies and those that specialize in the Internet.

Internet jobs within a traditional company include Internet strategists and administrators, Internet systems developers, Internet programmers and Internet or website operators.

Systems Developers

Systems developers design and write software. Typically, developers will be graduates with degrees in technical subjects such as computer science, mathematics or engineering. However, many big employers have graduate recruitment schemes where degree subject is less important than an ability to learn. On such schemes, graduates are taught the skills they need. The skills needed by developers include the ability to design solutions to problems and communicate these solutions to other developers and to users, and the technical skill to create these solutions. Software development can be extremely challenging and exciting.

Often, systems developers are employed to create software to support business goals, such as developing the organization's transaction processing system. Alternatively, systems developers may work in a software house, where the software they write is the product the organization sells. One of the fastest growing areas of software development is the games industry, with many universities now offering degrees in games development.

Other IS Careers

Other IS career opportunities include technical writing (creating technical manuals and user guides) and user interface design.

certification A process for testing skills and knowledge which results in a statement by the certifying authority that an individual is capable of performing a particular kind of job.

Often, the people filling IS roles have completed some form of certification. **Certification** is a process for testing skills and knowledge resulting in an endorsement by the certifying authority that an individual is capable of performing a particular job. Certification frequently involves specific, vendor-provided or vendor-endorsed coursework. Popular certification programs include Microsoft Certified Systems Engineer, Certified Information Systems Security Professional (CISSP), Oracle Certified Professional and many others.

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Summary

The use of information systems to add value to the organization is strongly influenced by organizational structure and the organization's attitude and ability to change. An organization is a formal collection of people and other resources established to accomplish a set of goals. The primary goal of a for-profit organization is to maximize shareholder value. Non-profit organizations include social groups, religious groups, universities and other organizations that do not have profit as the primary goal.

Organizations are systems with inputs, transformation mechanisms and outputs. Value-added processes increase the relative worth of the combined inputs on their way to becoming final outputs of the organization. The value chain is a series (chain) of activities that include (1) inbound logistics, (2) warehouse and storage, (3) production, (4) finished product storage, (5) outbound logistics, (6) marketing and sales, and (7) customer service.

Organizational structure refers to how organizational subunits relate to the overall organization. Several basic organizational structures include traditional, project, team and a virtual one. A virtual organizational structure employs individuals, groups or complete business units in geographically dispersed areas. These can involve people in different countries operating in different time zones and different cultures. Organizational change deals with how profit and non-profit organizations plan for, implement and handle change. Change can be caused by internal or external factors. According to the concept of organizational learning, organizations adapt to new conditions or alter practices over time.

Because information systems are so important, businesses need to be sure that improvements to existing systems, or completely new systems, help lower costs, increase profits, improve service or achieve a competitive advantage. Business process reengineering involves the radical redesign of business processes, organizational structures, information systems and values of the organization, to achieve a breakthrough in results. Continuous improvement to business processes can add value to products and services.

The extent to which technology is used throughout an organization can be a function of technology diffusion, infusion and acceptance. Technology diffusion is a measure of how widely technology is in place throughout an organization. Technology infusion is the extent to which technology permeates an area or department. User satisfaction with a computer system and the information it generates depends on the quality of the system and the resulting information.

Outsourcing involves contracting with outside professional services to meet specific business needs. This approach allows the company to focus more closely on its core business and to target its limited resources to meet strategic goals. Downsizing involves reducing the number of employees to reduce payroll costs; however, it can lead to unwanted side effects.

Competitive advantage is usually embodied in either a product or service that has the most added value to consumers and that is unavailable from the competition, or in an internal system that delivers benefits to a firm not enjoyed by its competition. The five-forces model explains factors that lead firms to seek competitive advantage: the rivalry among existing competitors, the threat of new market entrants, the threat of substitute products and services, the bargaining power of buyers, and the bargaining power of suppliers. Strategies to address these factors and to attain competitive advantage include cost leadership, differentiation, niche strategy, altering the industry structure, creating new products and services, improving existing product lines and services, and other strategies.

Cooperation between business managers and IS personnel is the key to unlocking the potential of any new or modified system. Information systems personnel typically work in an IS department. The chief information officer (CIO) employs an IS department's equipment and personnel to help the organization attain its goals. Systems analysts help users determine what outputs they need from the system and construct the plans needed to develop the necessary programs that produce these outputs. Systems analysts then work with one or more system developers to make sure that the appropriate programs are purchased, modified from existing programs or developed. The major responsibility of a computer programmer is to use the plans developed by the systems analyst to build or adapt one or more computer programs that produce the desired outputs.

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Computer operators are responsible for starting, stopping and correctly operating mainframe systems, networks, tape drives, disc devices, printers and so on. LAN administrators set up and manage the network hardware, software and security processes. Trained personnel are also needed to set up and manage a company's Internet site, including Internet strategists, Internet systems developers, Internet programmers and website operators. Information systems personnel can also support other functional departments or areas.

In addition to technical skills, IS personnel need skills in written and verbal communication, an understanding of organizations and the way they operate, and the ability to work with people (users). In general, IS personnel are charged with maintaining the broadest enterprise-wide perspective.

Self-Assessment Test

1. A minimum number of management layers results in a _____ organization structure.
2. Giving employees and their managers more responsibility and authority is known as _____.
3. Outsourcing computing resources to a third party is known as _____.
4. The radical redesign of business processes is known as _____.
5. _____ are information systems that an organization could not operate without.
6. Productivity equals _____.
7. If a firm can outperform its industry it has a _____.
8. _____ is the percentage of sales that a product or service has in relation to the total market.
9. An agreement between two or more companies is known as a _____.
10. The head of the information systems department is the _____.

Review Questions

1. What is the value chain?
2. Describe a virtual organization structure.
3. What is technology diffusion?
4. Describe on-demand computing.
5. Define continuous improvement and compare it with reengineering.
6. What is a support application?
7. Describe some of the ways in which IS can be evaluated.
8. Why would a company constantly develop new products?

Discussion Questions

1. Search on www.indeed.com or similar for some current information systems job opportunities for recent graduates. What skills are required? Which are the areas that interest you the most?
2. How would you help an employee cope with rapid organizational change?

Web Exercises

1. Search for information about the career of a CIO in a top organization. What skills must they have displayed at each stage?
2. Select an object on the desk in front of you. Sketch out its value chain. How were the raw materials that it is made of processed in a way that made them more valuable?

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Case One

Nailed it! Instagram and Cloud Computing Ensure Success

After her graduation in fashion, New Zealand born Anna Ross found herself working in Melbourne in the clothing industry, putting in 90 hour weeks for little money. 'I just couldn't do it anymore', she says. 'I decided I hated working in fashion.' Looking for a way out, she soon started designing and selling her own silver jewellery. With little start-up capital, it took two years but eventually she noticed a gap in the market and started to manufacture nail polish. 'Nail polish is something I have enjoyed from a young age', she says. 'It's fun and there is something almost cathartic about painting your nails.' The gap she spotted was an unmet demand for ethically produced, professional quality polish. 'I was really just testing the waters, but there turned out to be so much demand.' It took a year to perfect the product, which is vegan accredited, not tested on animals and made in Australia. She hired a chemist to help her get the colours right, without the need for any unpleasant chemicals. She called her brand Kester Black.

To market her product, Ms Ross used social media, carefully crafting an online presence that made her company seem much larger than it actually was. By publishing beautifully shot photos on Instagram, Ms Ross turned her brand into a lifestyle product that attracted customers who desired the lifestyle she portrayed in the photos. Nick Bez, a director at Australian market research group Mobium Group, told business reporter Kate Stanton that, 'Being savvy with social media and using the online environment to talk to a wider audience, means you can be a little brand and have a persona of something much bigger'.

In an interview with online magazine The Design Files, Ms Ross explained her use of cloud computing. When Kester Black got big enough that an additional employee was needed, she used the cloud to hand over responsibilities to her new administrator. Using Google Docs meant that they could both be working on the same document at the same time, from anywhere with an Internet connection. They used Gmail to communicate and Google calendar to coordinate manufacturing, accounting, pay reminders, marketing and event listings. They also used Dropbox for storage of all important documents that they both had to access. All of this software is available for free and offers basic functionality (although users must agree to having some of their data used for targeted advertising), yet it is powerful enough to have helped a young company get off the ground.

Questions

1. How can social media change people's perception of a company?
2. Are there any dangers with using social media as a platform for promotion?
3. What are some of the advantages of using Google Docs?
4. What other cloud-based services could small businesses benefit from?

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Case Two

Crosswords and Cybersecurity

In the 2014 film *The Imitation Game*, Alan Turing is shown recruiting code breakers by publishing a crossword in the *Daily Telegraph* with a message asking people who could solve it in ten minutes or fewer to get in touch. This is based on a real event. The actual crossword used was published on 13 January 1942 and gave readers 12 minutes not 10. Clues included 'when hammering take care not to hit this (5,4)' and 'those well brought up help these over stiles (4,4)'. It might seem like a strange way to apply for a job, but puzzles are still used by the intelligence community.

The current 'coding challenge' issued by the British security service MI5 is an image showing pink and blue lines with the instructions: 'Intelligence isn't always obvious and our engineers and analysts work hard to unlock it. There's a clue in the image file below, if you can find it'.

Opening the file in a text editor reveals a further Shakespearian sounding clue hidden in a comment: 'As I read, numbers I see. Twould be a shame not to count this art among the great texts of our time.' You can use a search engine to find the image using this line if you want to see it. The important part of the message seems to be 'count this art'. The challenge is to count the number of pink pixels followed by the number of blue pixels repeatedly. For that you'll need software that can pick apart an image file (you could try the 'png' package in the programming language R if you want to give it a go). Spoiler alert: to give it away, the numbers refer to letters using ASCII character coding (the American Standard Code for Information Interchange gives each character a numerical representation) to reveal the message, 'Congratulations, you solved the puzzle! Why don't you apply to join our team?' There's probably a new puzzle on the MI5 website by now if you want a fresh challenge.

In November 2016, UK Chancellor Philip Hammond said that the country must develop the capability to launch a counter cyber-attack. He said, 'The UK must strike back at hostile states in cyberspace and be capable of mounting sophisticated cyber-attacks of its own in place of military strikes'. With information technology having such a rapid pace of change, it is not an easy thing to recruit people with the skillset to deal with problems from technology that hasn't even been invented yet. 'Sometimes when you are screening [new recruits], you don't even know what kind of technology or problems they are going to be dealing with by the time they actually sit there and have the responsibility upon their shoulders', says Nadav Zafrir, an ex-member of Israel's famous Unit 8200 which runs electronic surveillance and other cyber activities.

Mr Zafrir left Unit 8200 in 2013 to found Team 8, which aims to build successful companies that can transform cybersecurity. Team 8 says that they work 'with innovative leaders and entrepreneurs to address the biggest problems in cybersecurity and create differentiated, independent leaders in the cybersecurity market. Our mission is to initiate and build long-term, innovative cybersecurity companies that can truly make the world a safer place and assist enterprises in coping with the most daunting cyber challenges'. The skills that Team 8 are developing are exactly what the UK will need if it is to develop its cyber-war capabilities.

'We literally had to redefine [a] control hierarchy as otherwise we wouldn't be able to create the magic that these kids can bring with them. I think a lot of them are more agile, better prepared, speak languages that we didn't even know existed in terms of coding and stuff like that', Mr Zafrir says. 'We had to adapt not only the recruiting process, but also the day-to-day control systems in many ways in order to create this magic.'

And the answers to Alan Turing's puzzle? 'Right nail' and 'Lame dogs'. Don't be too surprised if you didn't get them. Apparently, Telegraph readers in the 1940s had very different ideas about how crossword clues work from what we have today.

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Questions

1. How could children at school be encouraged to consider a career in cybersecurity?
2. How could job recruitment be made to be 'future proof'?
3. Should governments rely on private firms for their nation's cybersecurity?
4. How could this same approach be used by organizations to hire IT professionals?

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Case Three

Raining Wine from the Cloud

Baacco is a search engine which helps customers find and buy premium yet affordable wines. It describes itself as a 'cloud first business' and uses over 20 cloud-based tools to reach customers in the UK, France, Germany, Spain and Belgium. The idea came when founder Tai Alegbe was trying to find a wine he had seen on a London restaurant's menu. He felt it should be possible to buy it somewhere but couldn't find it in a store. Sensing a challenge, he set about searching for a UK distributor who supplied it and found that it was difficult but not impossible to buy. He then realized there was a gap in the marketplace, so he and four friends founded Baacco, 'the ultimate new way to search, discover and shop for top-quality and affordable wines; whilst connecting with specialist wine merchants'. The Baacco website allows customers to search for wines from distributors and new sellers who are not represented in the supermarket chains and high street wine stores. Crucially, Baacco facilitates communication between sellers and customers to create an open discussion between wine experts and consumers who don't really understand grapes yet know what they like.

Baacco uses the cloud to process payments in different regions and currencies and to analyze their customers' experience. For instance, they use Crazy Egg to see how people are using their website. Crazy Egg can create maps of the most visited pages in a website from customer clicks, and a map of how far down a web page customers tend to scroll. The idea is to ferret out the parts of the site that aren't viewed so that they can be changed. Baacco uses Mixpanel to do something similar for users of their mobile app. Mixpanel will help predict which app users are most likely to use to make a purchase and when.

These tools have been essential. 'They allow us to build, measure and learn about our customers far quicker than any other method', Mr Alegbe says. One warning raised by Crazy Egg was that customers in France and Spain were getting to the checkout and then leaving the website. Investigating this, Baacco found that those customers were being put off by unfamiliar payment logos.

The cloud tools helped the business to grow. 'We scaled up our business by gaining an early understanding of which marketing channels had proven to be most successful, and then making sure the unit economics for that channel made sense for the business', said Mr Alegbe. They also helped the business appear much bigger to customers than it really was, which is a recurring theme in case studies of small businesses who access the cloud. 'The cloud provides a more affordable, flexible and scaleable platform that greatly enhances small retailers' opportunities to manage, share and control its data – and function like a globally integrated enterprise', says Vish Ganapathy, vice president and chief technology officer at IBM.

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Questions

1. What were the advantages to Baacco in using cloud computing?
2. How would knowing where people stop scrolling down a web page help to redesign that page?
3. How could a company decide between two possible changes they are considering making to their website?
4. Investigate and list some other cloud services Baacco could potentially use.

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World Views Case

ABAZONG Training and Consulting (Pty) Ltd Helps Organizations and Users With Information Security Consultation and Training Awareness

Paul Abanda
Damelin College, South Africa

ABAZONG Training and Consulting is run by individuals that collectively have decades of experience working within businesses. Their team focuses on helping organizations implement working solutions to information security management standard. Their staff ensure that there is a business-wide solution to organizational security data, which can take the form of physical and logical infrastructures.

The organization has established relationships in order to support businesses and enable them to offer services in the areas of information security corporate training, penetration testing and information security auditing, software project development and security management. Their staff members are knowledge leaders in the industry and are among the most active Sub Sahara African participants in industry conferences, blogs and podcasts.

Furthermore, the team has addressed security problems in South Africa and several other African countries, and they have a good understanding of information security and the standards that govern security. The ISO 27001 information security management systems audit (ISMSA) standard considers every risk critical in identifying potential dangers and disruption in the quality, quantity and distribution or relevance of data that can put a business at risk.

ISO 27001 improves business reliability and demonstrates the integrity of data, systems and the obligation to information security. This helps to transform organizational culture, opening up new venture opportunities with security conscious customers and thereby improving employee ethics and strengthening the notion of workplace confidentiality.

The precision of ABAZONG Training and Consulting's technical competence is guaranteed to ensure that businesses operate at optimal levels in their requirements, which are data loss prevention, vulnerability management, logical access, anti-virus protection, system integration, software integration, or any other specialized IT security services. Their dedicated technical professionals ensure that businesses receive the most appropriate solution that is aligned to meet the objectives of ISO 27001's information security management systems standard.

Questions

1. What are the advantages and disadvantages of creating awareness of the ISO 27001 information security management systems standard?
2. Identify the dangers that are associated with lack of training on the protection of organizational information systems.
3. What is a code of ethics and why should employees be ethical?
4. How does ABAZONG Training and Consulting assist organization security awareness?
5. Why should IS professionals become certified in ISOs?

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